LITTLETON PUBLIC SCHOOLS
LIBRARY OF THINGS · ITEM GUIDE

2 kits

Makedo Cardboard Construction Kit

WHAT'S IN THE KIT

- Makedo screws and connectors
- Cardboard saws (kid-safe)
- Screwdriver tools per kit

QUICK-START GUIDE

1. Collect cardboard for a week before checkout — you'll need more than you think.
2. Demo the safe-saw technique: saw away from the holding hand.
3. Students punch, screw, and connect — no tape or glue.
4. Unscrew everything at the end; the connectors are all reusable.

FULL LESSONS IN THIS GUIDE

PRO TIP: *Combine with Canary Knives for older students who need cleaner, faster cuts.*

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LESSON · GRADES K-2

Box City

LEARNING OUTCOMES

- Students will use construction tools safely and purposefully
- Students will collaborate on a large shared structure
- Students will describe spatial relationships in a built environment

MATERIALS

- Makedo kits (screws, safe-saws, tools)
- A large stash of cardboard boxes (collect for a week)
- Floor plan area with tape 'streets'
- Toy figures for the finished city

INSTRUCTIONS

1. Tool school first: demo the safe-saw motion and screw-driving on scrap.
2. Assign zones: houses, shops, tunnels — each small group owns a build.
3. Build sessions with a repack ritual at the end of each.
4. Connect builds along the tape streets into one city.
5. Walk toy figures through while narrating: over, under, next to, through.
6. Unscrew everything at the end — the connectors' reusability is the closing lesson.

ACTIVITY

> Box City — the class engineers a connected cardboard city with real tools, then narrates journeys through it. Collaboration and spatial language at architectural scale.

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LESSON · GRADES 3-5

Cardboard Arcade

LEARNING OUTCOMES

- Students will engineer a working game with a moving mechanism
- Students will test and refine for real users
- Students will run a service experience for an audience

MATERIALS

- Makedo kits + Canary knives for cleaner cuts
- Cardboard boxes of all sizes
- Balls, marbles, rubber bands for game mechanics
- Ticket/prize supplies for arcade day

INSTRUCTIONS

1. Watch or retell the Caine's Arcade story as the hook.
2. Teams design a game with at least one moving mechanism (flap, chute, lever, catapult).
3. Build across 2-3 sessions with a mid-point playtest.
4. Playtest fixes: too easy, too hard, or breaks after five plays all require redesign.
5. Arcade day: run games for another class or families; team members rotate roles.
6. Debrief with real user data: which game had the longest line, and why?

ACTIVITY

> Cardboard Arcade — engineer games sturdy enough for strangers to play. Arcade day converts engineering into hospitality: rules, repairs, and lines out the door.

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LESSON · GRADES 6-8

Wearable Cardboard Tech

LEARNING OUTCOMES

- Students will design wearable structures with moving parts
- Students will prototype rapidly with cardboard construction systems
- Students will integrate electronic interactivity into a physical build (optional)

MATERIALS

- Makedo kits + Canary knives
- Large cardboard sheets
- Brads, string, elastic for articulation
- Optional: Makey Makey or micro:bit for interactive upgrades

INSTRUCTIONS

1. Design brief: wearable armor, helmet, or product prototype with at least one articulated part.
2. Pattern first: trace body measurements onto cardboard before cutting.
3. Build with articulation: hinged joints, sliding visors, grabber extensions.
4. Wearability test: put it on, move, sit, reach — fix what binds.
5. Optional tech layer: a micro:bit heartbeat display or Makey Makey sound trigger.
6. Runway demo with a 30-second design-decisions narration.

ACTIVITY

Wearable Cardboard Tech — armor and prototypes that must survive being worn. Articulated joints teach mechanism design; the optional micro:bit layer turns costume into product.